-REPORT-

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1. Experiment 3:

Listing 5

```
# Listing 5 - print a message and an integer

.data
msg:    .asciiz "Usage (kWh): "
nl:     .asciiz "\n"

.text
.globl main
main:
    addi $t0, $zero, 150     # Store usage value 150 in $t0

    li   $v0, 4              # service 4: print string
    la   $a0, msg            # argument: address of msg
    syscall                 # print "Usage (kWh): "

    li   $v0, 1              # service 1: print integer
    add  $a0, $t0, $zero     # argument: copy value from $t0 to
$a0
    syscall                 # print integer 150

    li   $v0, 4              # service 4: print string
    la   $a0, nl             # argument: address of newline
string
    syscall                 # print a new line

    li   $v0, 10             # service 10: exit program
    syscall                 # terminate program
```

PREDICT – text in the Run I/O pane:

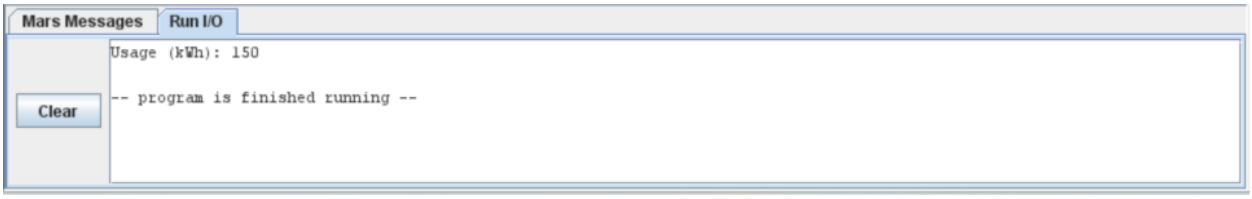
```
Usage (kWh): 150
```

OBSERVE – run the program. Then open Settings → Show Labels Window (symbol table) and read the addresses of the two labels.

Table 11. Label addresses

Label	Address (hex)
`msg`	0x10010000
`nl`	0x1001000e

Screenshot of the Run I/O pane after the program has run:



Listing 6

```
# Listing 6 - energy used between two readings

.data
prompt1:  .asciiz "Enter previous reading (kWh): "
prompt2:  .asciiz "Enter current reading (kWh): "
label:    .asciiz "Usage (kWh): "
newline:  .asciiz "\n"

.text
.globl main
main:
   # Register plan:
   # $s0 = previous reading
   # $s1 = current reading
   # $t0 = usage (current - previous)

   # Read the previous reading into $s0
   li   $v0, 4
   la   $a0, prompt1
   syscall              # Print previous reading prompt

   li   $v0, 5
   syscall              # Read previous reading
   add  $s0, $v0, $zero    # Store previous reading in $s0

   # Read the current reading into $s1
```

```
    li   $v0, 4
    la   $a0, prompt2
    syscall             # Print current reading prompt

    li   $v0, 5
    syscall             # Read current reading
    add  $s1, $v0, $zero    # Store current reading in $s1

    # Calculate usage = current - previous
    sub  $t0, $s1, $s0      # Store usage in $t0

    # Print label and usage
    li   $v0, 4
    la   $a0, label
    syscall             # Print usage label

    li   $v0, 1
    add  $a0, $t0, $zero    # Copy usage to $a0
    syscall             # Print usage integer

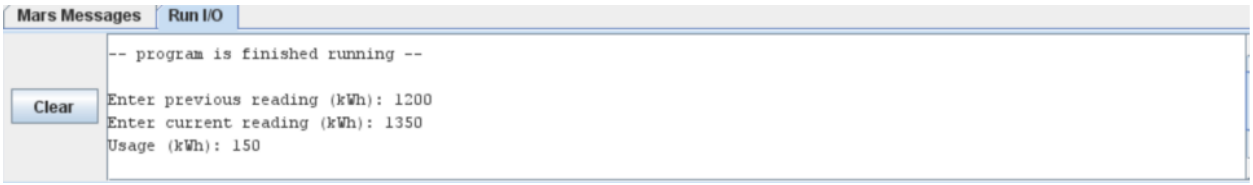
    # Print a newline
    li   $v0, 4
    la   $a0, newline
    syscall             # Print newline

    # Exit program
    li   $v0, 10
    syscall
```

Check your result

Run the program and type `1200` and `1350` when asked. The last line in the Run I/O pane must be `Usage (kWh): 150`. Type each number in the Run I/O pane and press Enter.

Screenshot of the Run I/O pane for this test:



Extend your program so that it reads four readings in order (reading0 to reading3) and then prints the usage of each month and the total. Start from the `.data` section below and keep the register plan of Listing 3 (`$s0–$s3` for the readings, `$t0–$t3` for the results). The output must have this form:

```
Month 1: <usage1>
Month 2: <usage2>
Month 3: <usage3>
Total:   <total>
```

Listing 7 (data section)

```
# Listing 7 - data section

.data
prompt: .asciiz "Enter reading (kWh): "  # Prompt for user input
m1:     .asciiz "Month 1: "              # Label for month 1
m2:     .asciiz "Month 2: "              # Label for month 2
m3:     .asciiz "Month 3: "              # Label for month 3
tot:    .asciiz "Total: "              # Label for total usage
nl:     .asciiz "\n"                 # Newline character
```

First test the program with the readings 1200, 1350, 1475 and 1640; the four results must be 150, 125, 165 and 440. Then take D, the last digit of your student ID, and look up your own readings in Table 12. PREDICT the four results by hand, run the program with your readings, and record what you observe.

Table 12. Personal readings (D = last digit of the student ID)

D	reading0	reading1	reading2	reading3
0	1000	1120	1220	1360
1	1020	1143	1248	1390
2	1040	1166	1276	1420
3	1060	1189	1304	1450
4	1080	1212	1332	1480
5	1100	1235	1360	1510
6	1120	1258	1388	1540
7	1140	1281	1416	1570
8	1160	1304	1444	1600

9	1180	1327	1472	1630

Table 13. Results for your own readings

My D	Result	Month 1	Month 2	Month 3	Total
	PREDICT	147	145	158	450
	OBSERVE	147	145	158	450

Screenshot of the Run I/O pane with your own readings:

